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DEMONSTRATE AN UNDERSTANDING OF COMPUTER NETWORK COMMUNICATION

US 14942

SECTION A: FORMATIVE ASSESSMENT

Question 1

1. Bandwidth refers to the data carrying capacity of a channel or medium. Higher Bandwidth communication channels support higher data rates.
2. Radiation refers to the leakage of signal from the medium due to undesirable electrical characteristics of the medium.
3. Noise absorption is the susceptibility of the media to the external electrical noise that can cause distortion of data signal
4. Attenuation is the loss of energy as the signal proper gates outwards. The amount of energy lost depends on frequency,. Radiations and physical characteristics of media contribute to attenuation.

Question 2

1. File check sequence errors :Nodes that transmit frames append an FCS number which let's receiving device determine if the packet is complete and correct upon arrival.
2. Alignment errors : the length of a frame in bits should always be evenly divisible by 8,if the frame doesn't meet the criterion, it has an Alignment error
3. Collision and late Collision occur when more than one device tries to use the network at the same time.
4. Discards: a switch can discard packets that arrive tagged for a specific virtual LAN on a port not configured for that VLAN . Discard force TCP applications to resend packets which increases application latency,performance problems for UDP applications as well.
5. Unknown protocol errors: A switch or router can receive a packet whose meaning isn't understood due to receiving device having a particular protocol disabled when it is in fact needed.
6. Size errors: frames that are too short or too long or giant generate errors
7. Performance degradation is the issue involving loss of speed and data integrity due to poor transmission.
8. Host identification
9. Security issues involve maintaining network integrity preventing unauthorized users from infiltrating the system and protecting the network denial of service attacks

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10. Workstation or server: the data passing through the components may be delayed or blocked which impacts network Performance

Question 3

The data link layer is the second layer of the seven layer of the OSI model of computer networking. The data link layer's function is to transfer data between adjacent network nodes in a wide area network or between nodes on the same local area network segment. The data link layer provides the functional and procedural means to transfer data between network entities and detect and correct errors that may occur in the physical layer.

Question 4

1. Connectionless communication: IP ,a data packet can travel from a sender to a recipient without the recipient having to send an acknowledgement.
2. Host addressing: every host in the network must have a unique address that determines where it is.
3. Message forwarding: since networks are partitioned into sub networks and connect to other networks for wide-area communications, networks use specialized hosts to forward packet ls between networks.
4. Routing: routing focuses on the end -to-end logic of forwarding data.

Question 5

Administers can us SNMP monitor and manage aspects of their network by

1. Gathering information on how much bandwidth is being used on the network
2. Active polling the network devices to ask for a status at specified intervals
3. Notifying the admin by text message of a device failure
4. Collecting error reports which can be used for troubleshooting
5. Emailing an alert when the server reaches a specified low disk space level

Question 6

1. CPU utilization and operating temperature, it informs you about CPU utilization, temperature and performance. It helps you to identify possible problems before they become critical
2. Memory usage, it is easy to keep track of your memory usage and avoid bottlenecks in your network with it.
3. Bandwidth requirements ,you can measure available Bandwidth and real-time data ,detect overloads in an instant and prevent network strain

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4. Free hand disk space, monitoring your disk space keeps the overview of your memory capacities.
5. System in, Computer monitoring Gathers all the information about your IT infrastructure.

Question 7

1. Processor utilization report displays the processor utilization if a particular network device over a period of time
2. Memory utilization report displays percentages of free memory on a particular device on a particular network device over a period of time
3. Interface traffic volume report displays the rate of inbound and outbound traffic that goes through the selected port or interface over time
4. Interface error packet analysis report displays the percentage of error packets or discarded packets for the selected port or Interface
5. Interface packet analysis report displays the types of packets that transverse the selected port or Interface
6. Custom performance report displays performance counter values

Question 8

Error 800 occurs when you're trying to establish a new connection to a VPN server. It indicates that messages being sent by the VPN client are failing to reach the server .

1. The client device lost connection to its local network
2. The user specified an invalid name or address for the VPN server
3. A network firewall is blocking the VPN traffic